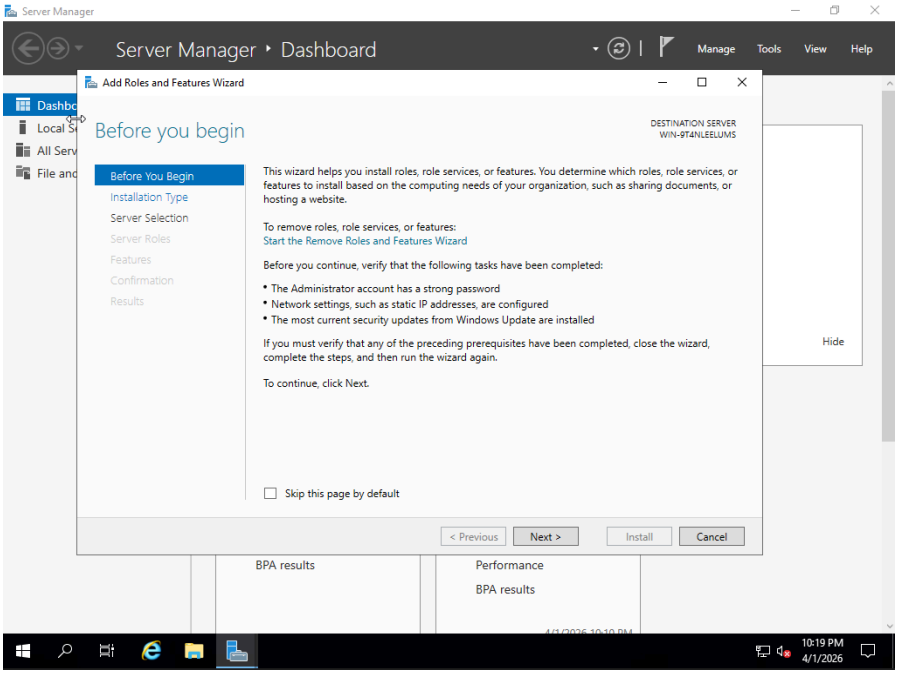
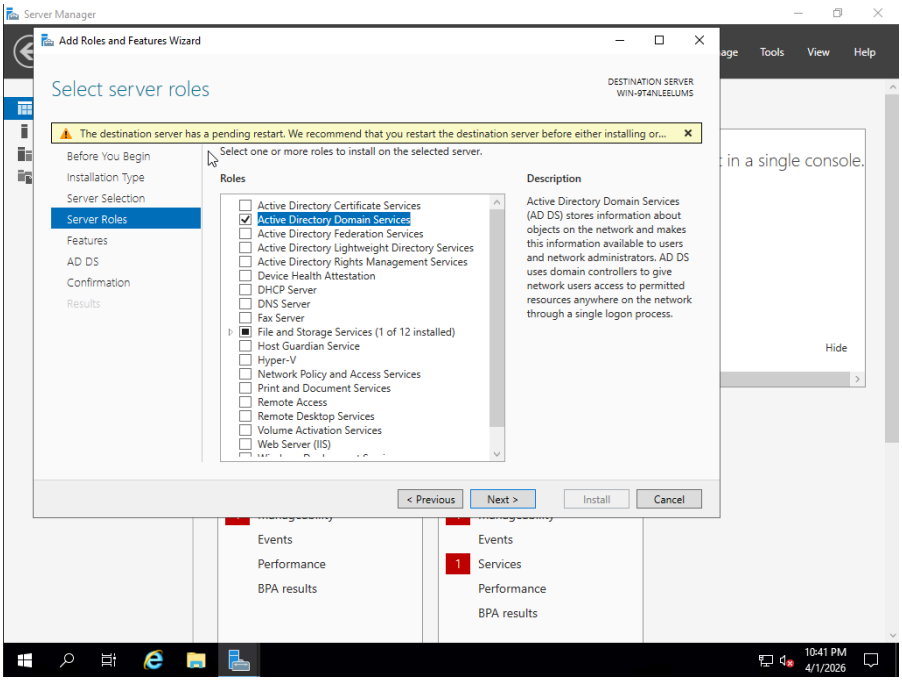
Active Directory Administration Lab (Windows Server)

1. Installing Active Directory Domain Services

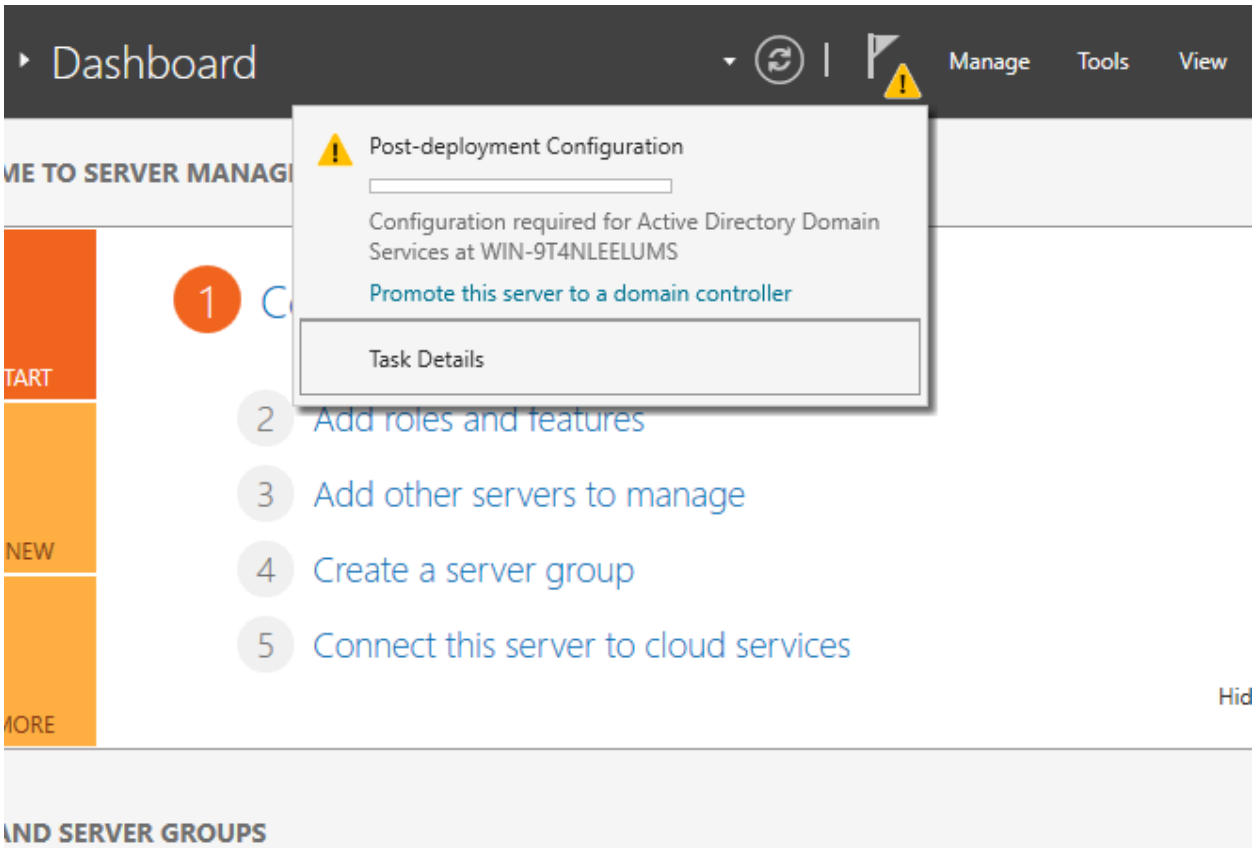
- Opened Server Manager
- Selected Manage → Add Roles and Features



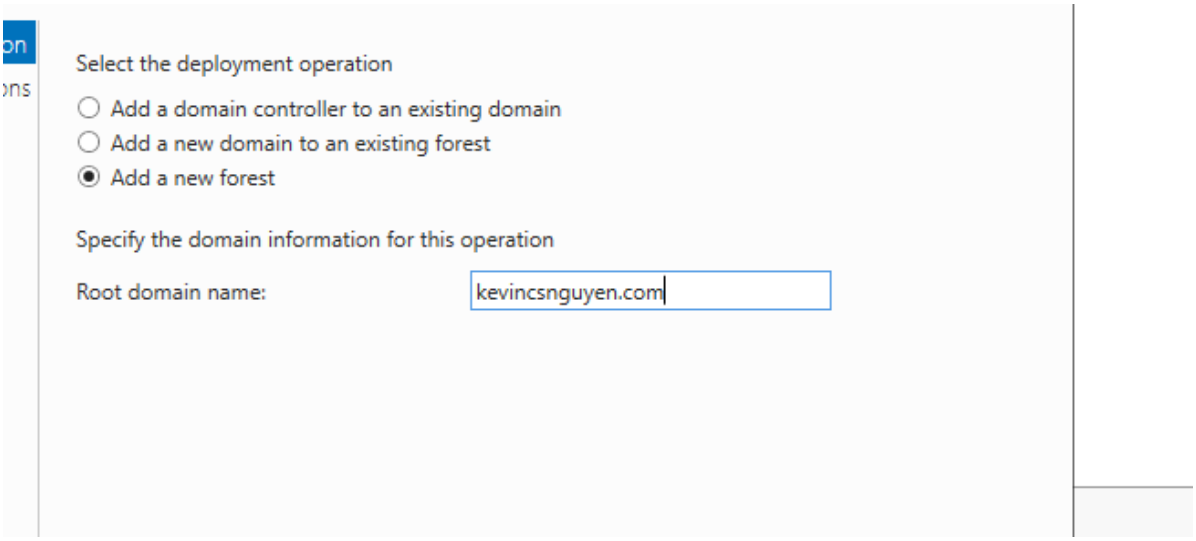
- Choose Active Directory Domain Services from the server roles list
- Completed the installation wizard

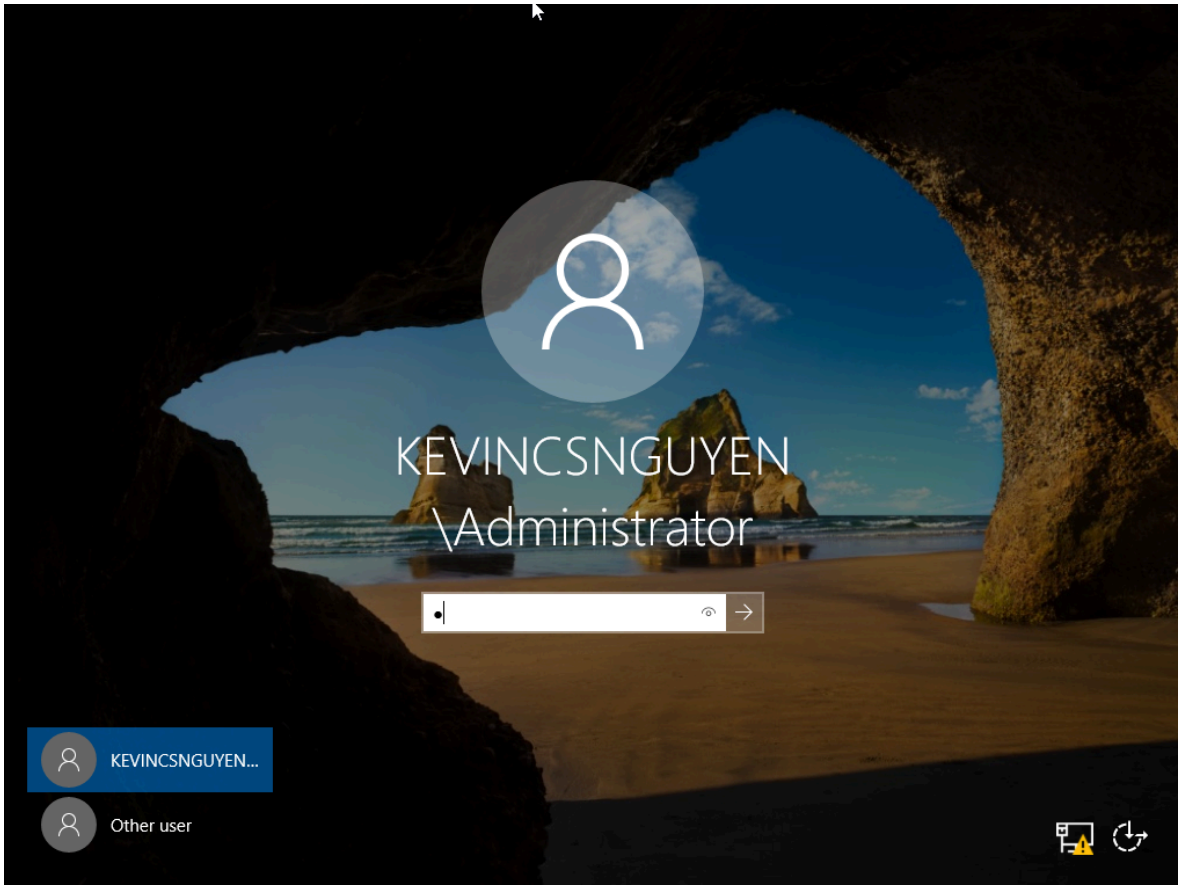


After installation, the server was **promoted to a Domain Controller**.

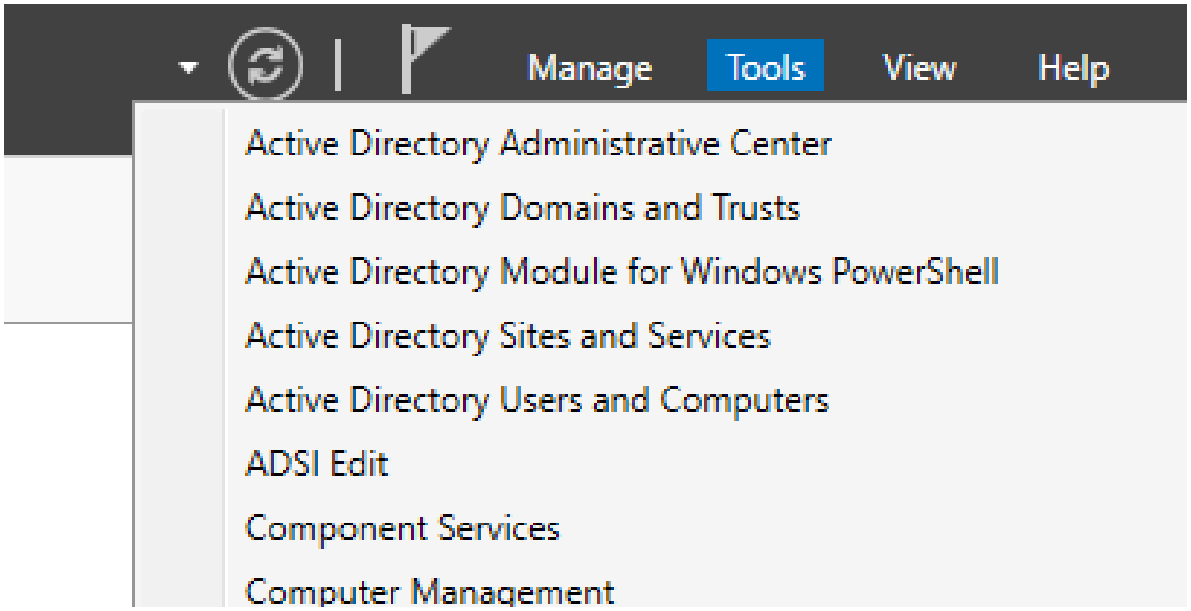


- Domain name used: kevincsnguyen.com
- Default configuration settings were applied for DNS and forest level.
- After completing the setup, the server automatically restarted and became the Domain Controller for the new domain.



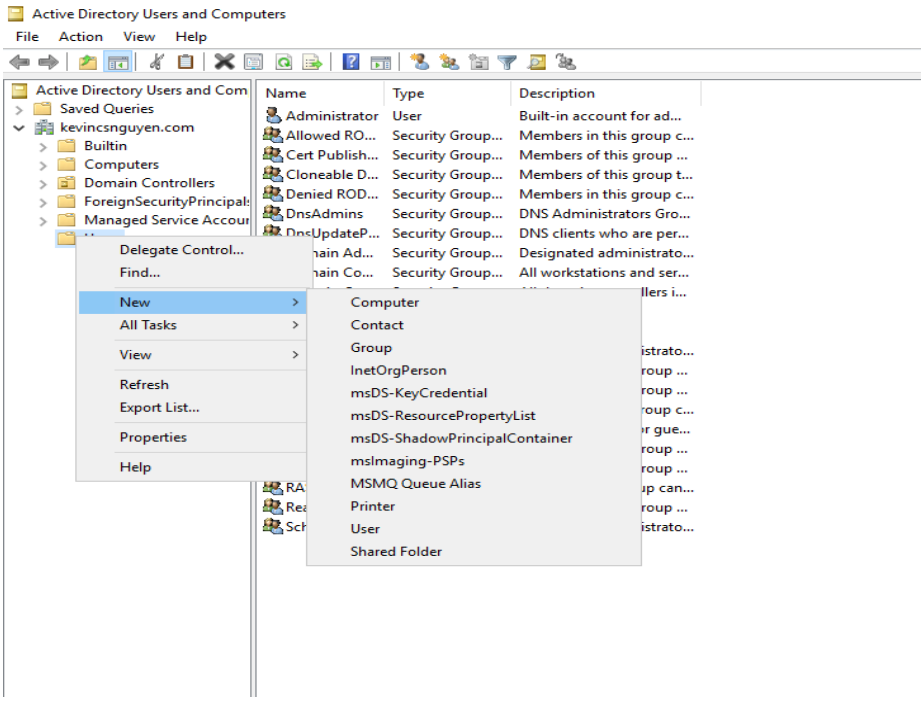


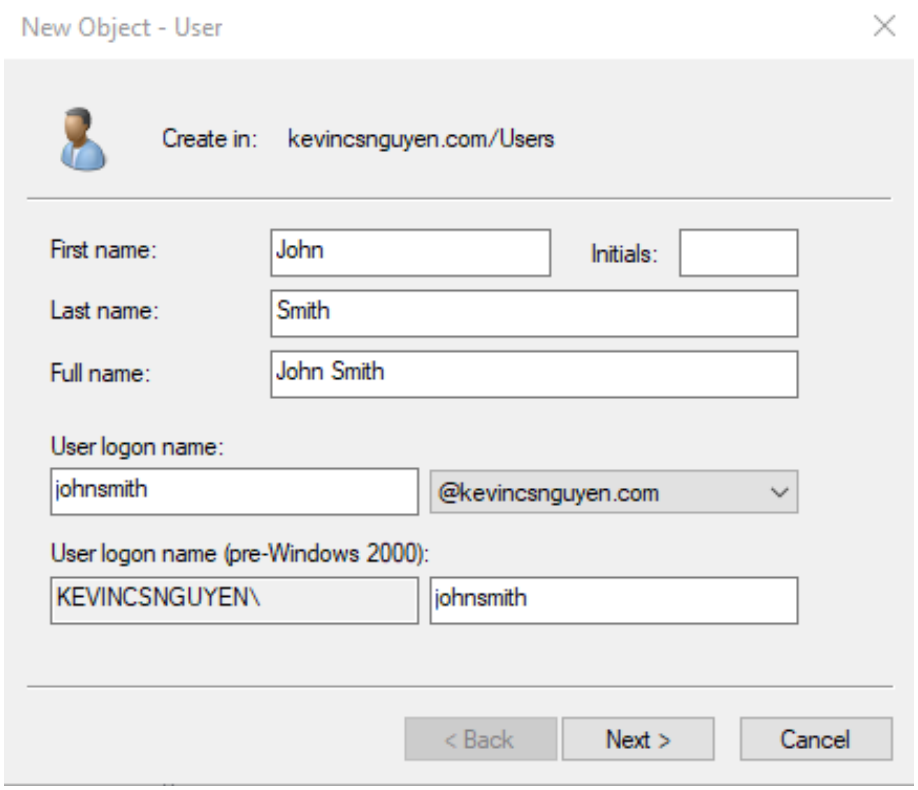
Administrative tools such as **Active Directory Users and Computers** were then available for managing domain objects.

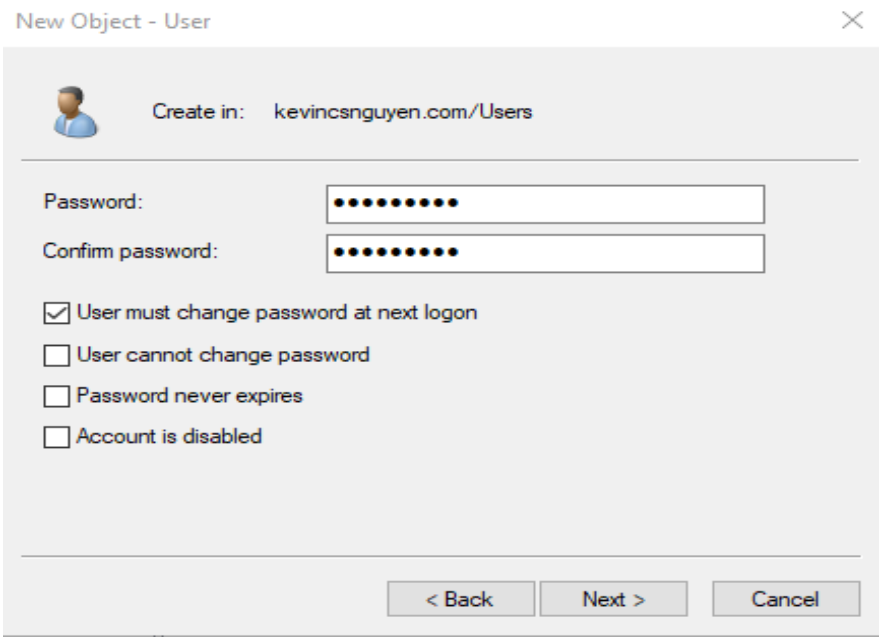


2. Creating and Managing User Accounts

- Opened **Active Directory Users and Computers**
- Navigated to the **Users container**
- Right-clicked → **New** → **User**
- Created a user profile with a username and password.

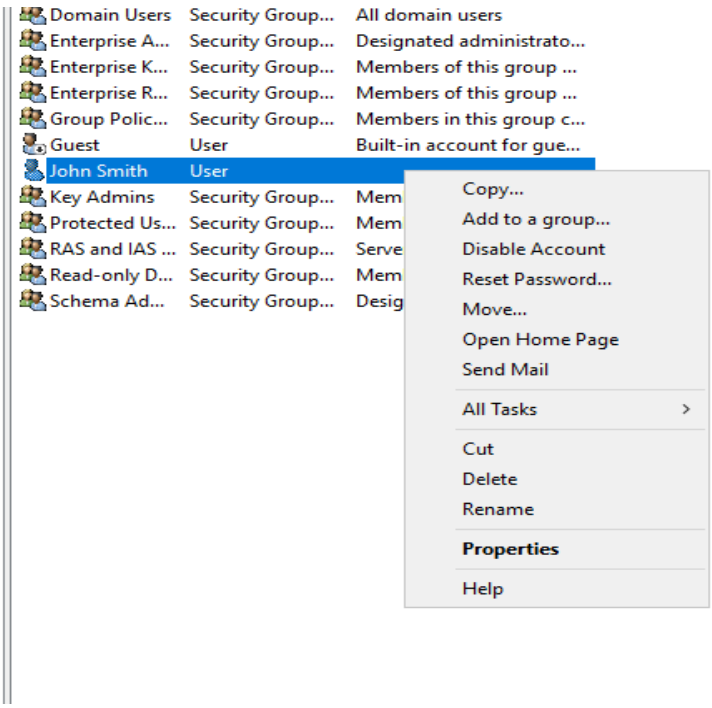






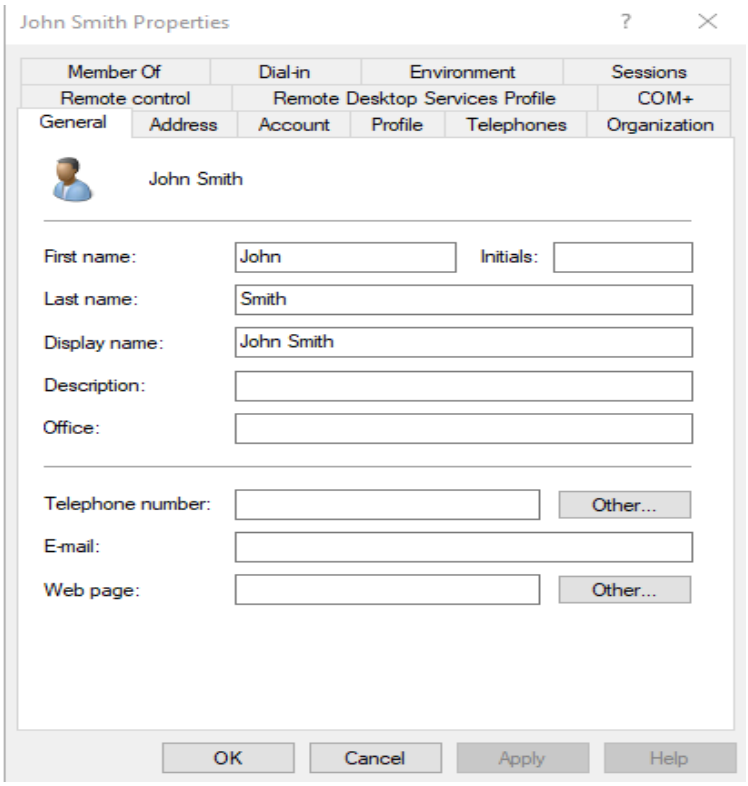
Active Directory allows administrators to perform many account management tasks including:

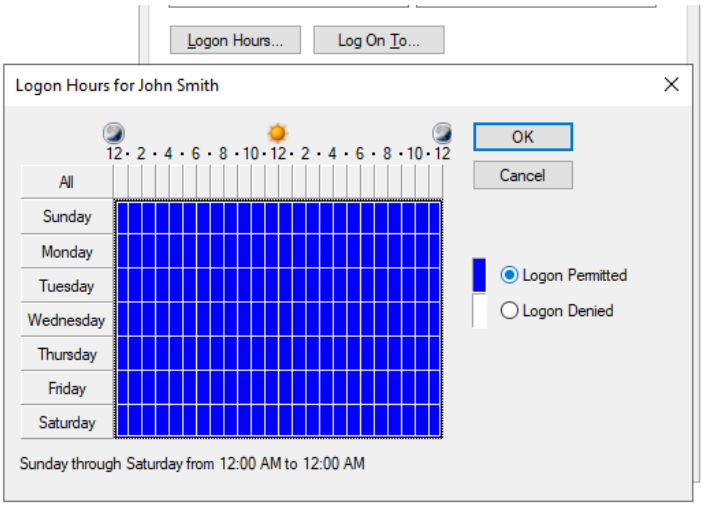
- Resetting user passwords
- Disabling or enabling accounts
- Adding users to security groups
- Managing account properties



Additional configuration options were explored by opening the **user properties panel**, where administrators can configure settings such as:

- Logon hours restrictions
- Account expiration
- Group memberships
- Profile settings





The user account was also verified using **Command Prompt** with the following command:

net user johnsmith /domain

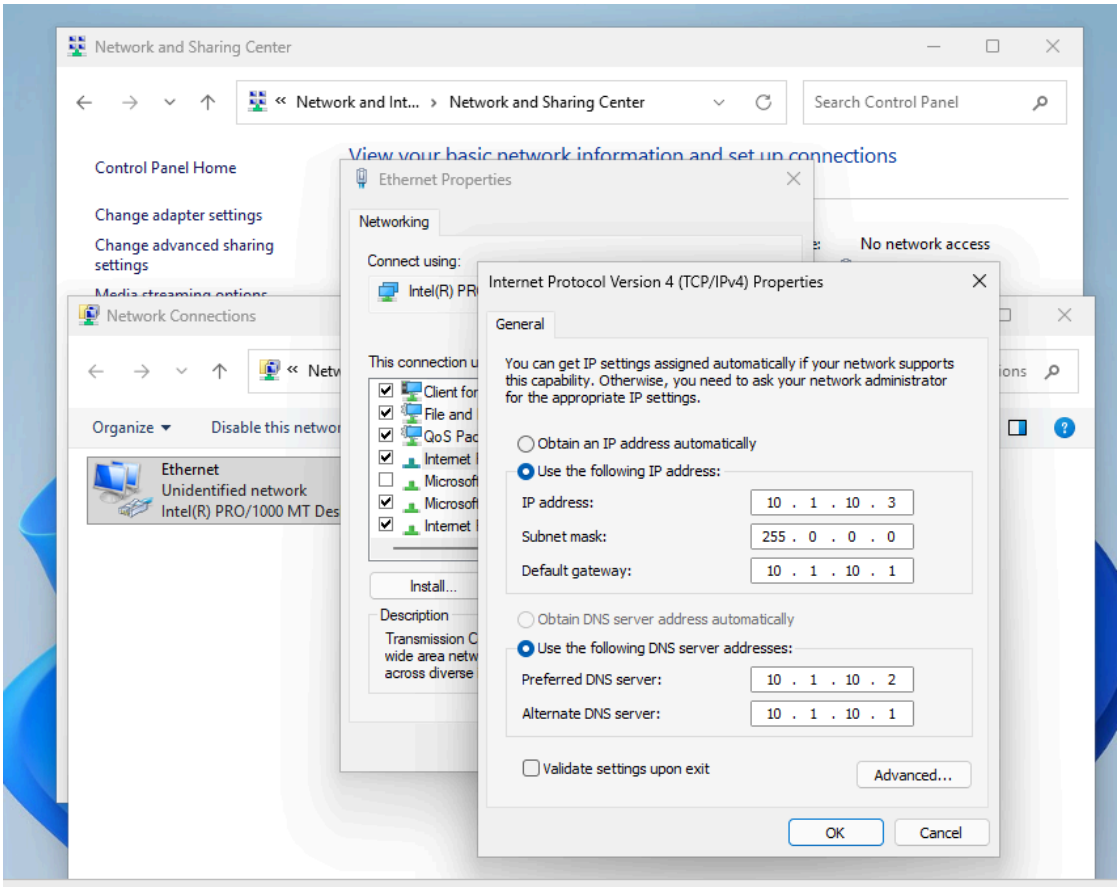
```
C:\Users\Administrator>net user johnsmith /domain
User name                    johnsmith
Full Name                    John Smith
Comment
User's comment
Country/region code          000 (System Default)
Account active               Yes
Account expires              Never

Password last set            4/2/2026 8:15:22 PM
Password expires             5/14/2026 8:15:22 PM
Password changeable          4/3/2026 8:15:22 PM
Password required            Yes
User may change password     Yes

Workstations allowed         All
Logon script
User profile
Home directory
Last logon                   Never

Logon hours allowed          All

Local Group Memberships
Global Group memberships     *Domain Users
The command completed successfully.
```

3. Joining a Workstation to the Domain

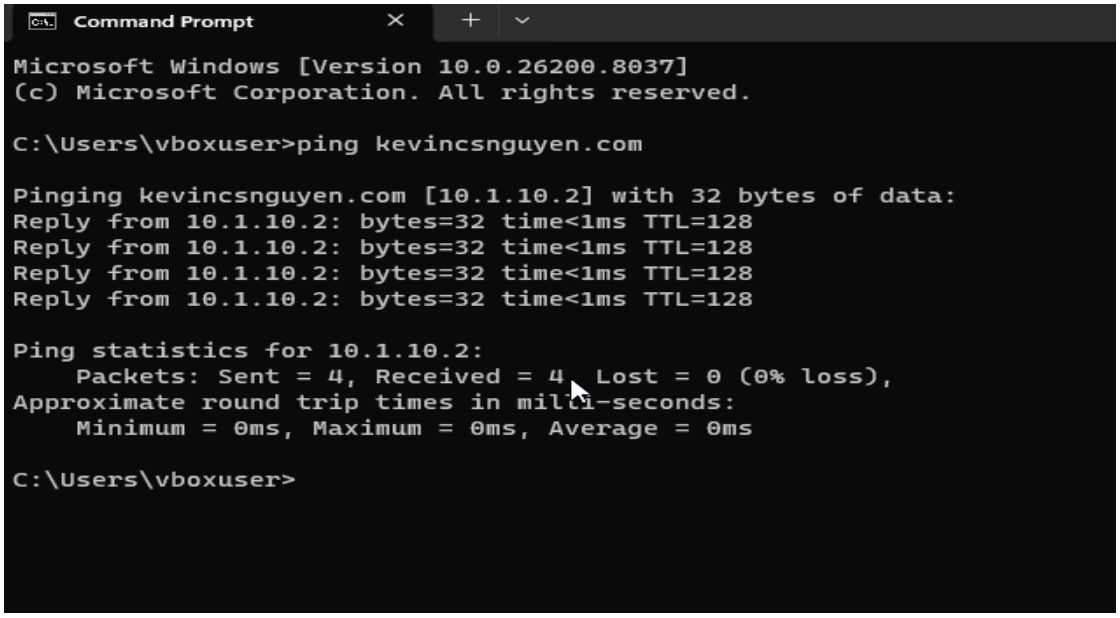
A **Windows 11 Pro virtual machine** was used to simulate a user workstation.

Note: Windows 11 **Pro edition is required** to join a domain. The Home edition does not support domain joining.

Configured the workstation's **DNS server** to point to the **Domain Controller**.



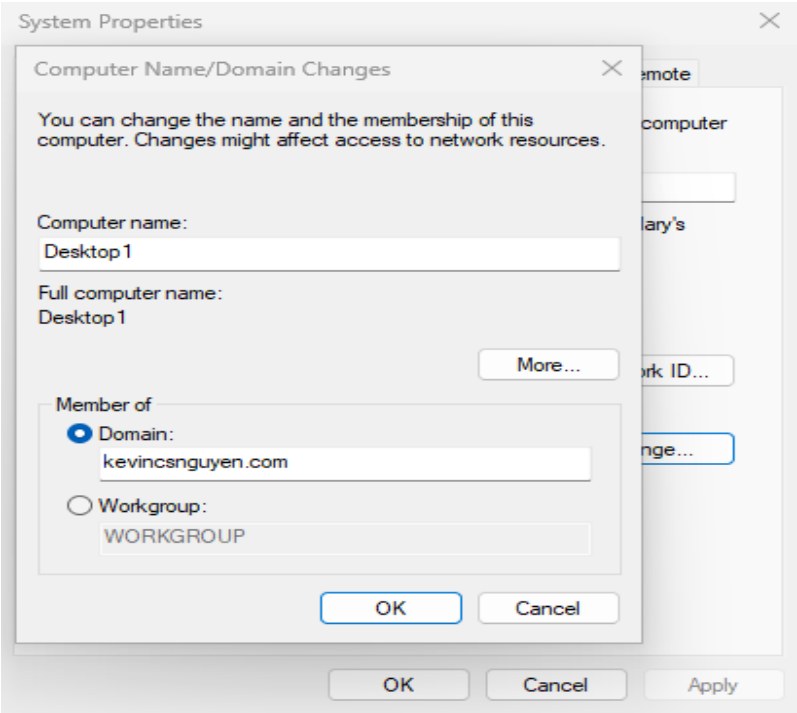
Verified network connectivity using the **ping command**.
Example: ping kevincsnguyen.com



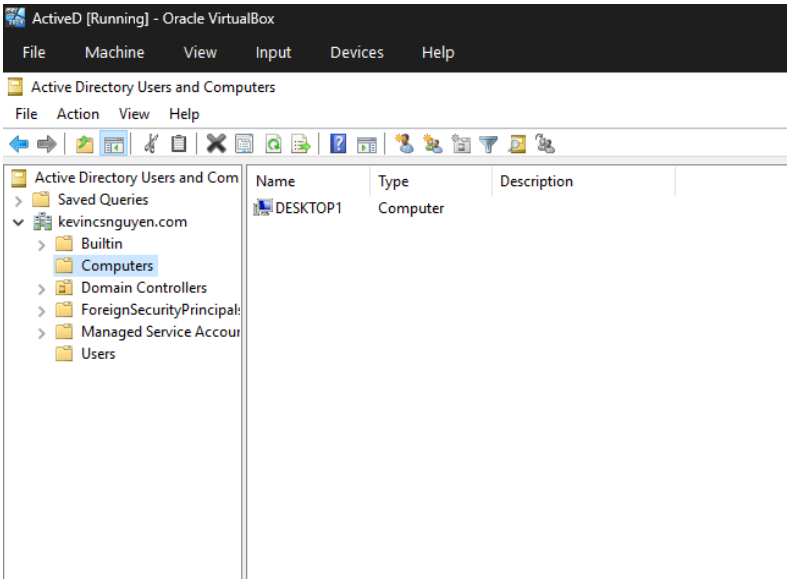
John Smith PC successfully pinged to my main server.

After confirming connectivity:

- Opened **System Properties**
- Selected **Change settings**
- Joined the workstation to the domain **kevincsnguyen.com**

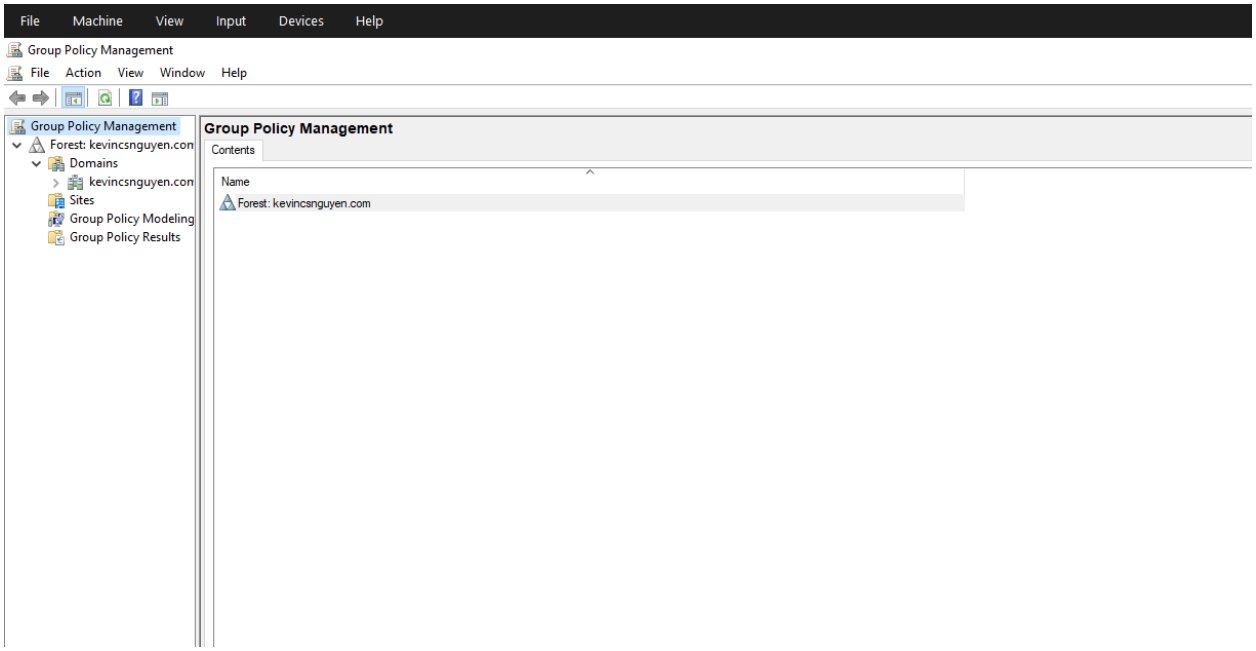


I can see that DESKTOP1 which is from John Smith was successfully joined when I look at the Active Directory.



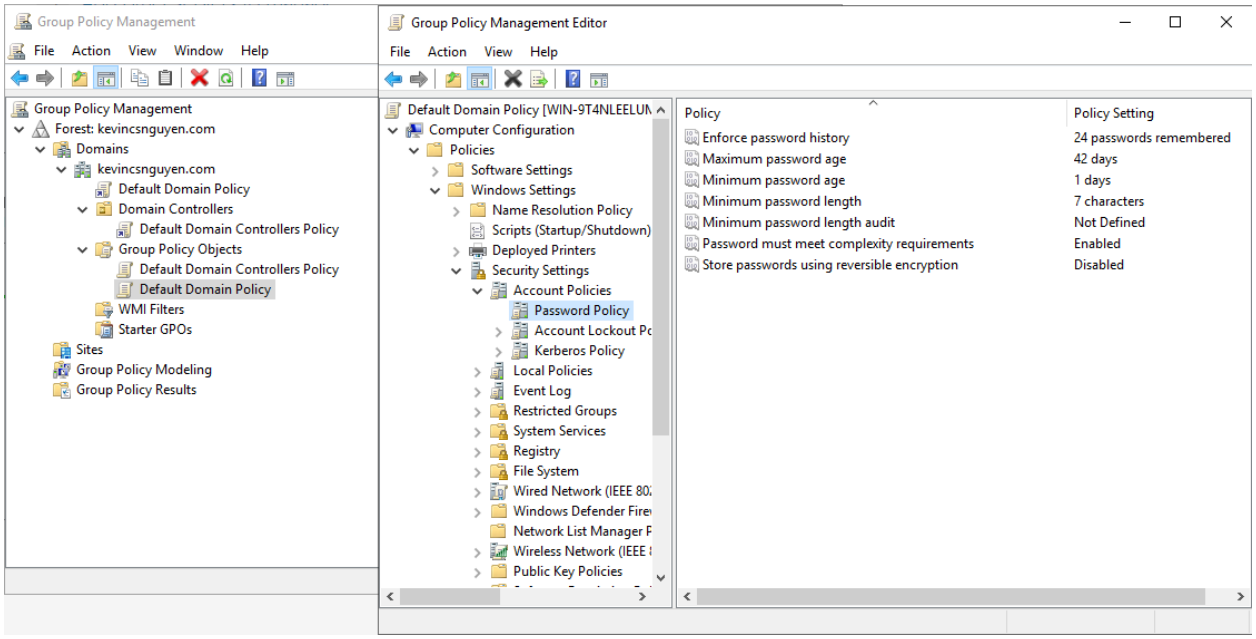
4. **Configuring Group Policy**

- Opened **Group Policy Management**
- Domain → Group Policy Objects → Default Domain Policy

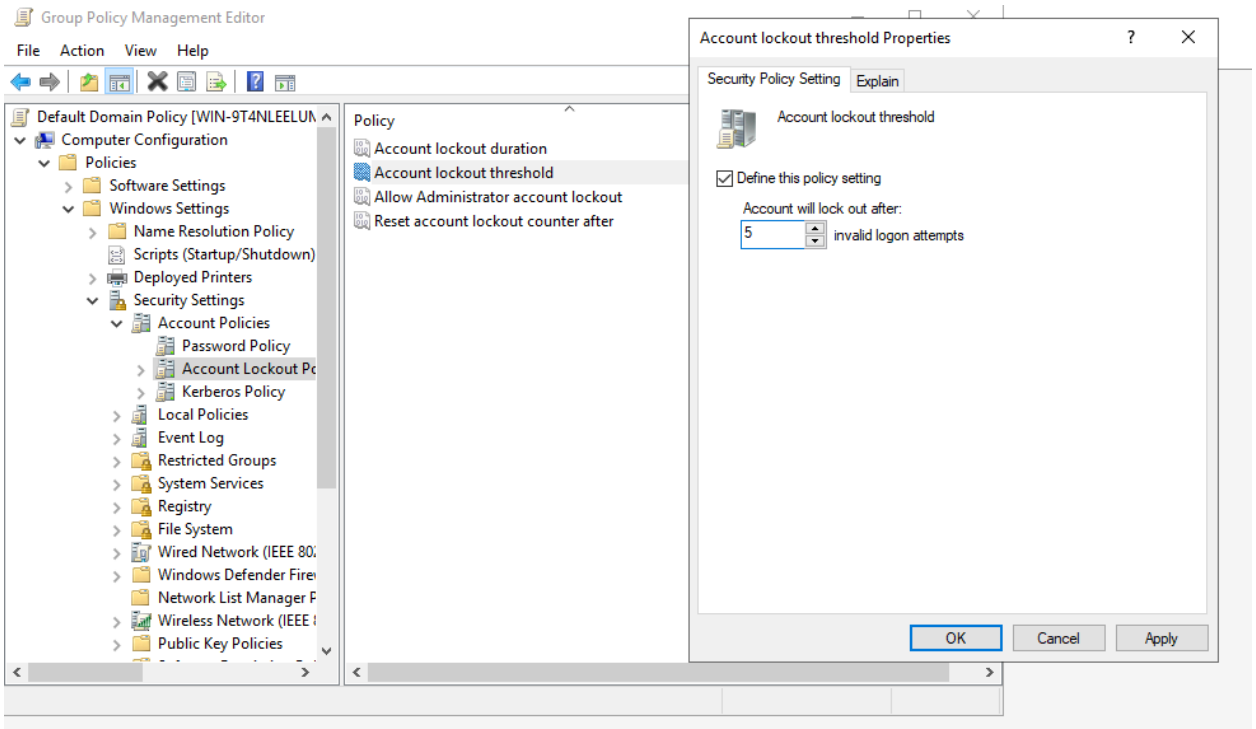


Let's say I want to change the company policy about workers passwords. I want everybody's password to expire and be required to be changed every 90 days for security purposes. I can do that with group policy management.

Example: From my domain kevincsnguyen -> Group Policy Objects -> Default Domain Policy (Edit) -> Window Settings -> Password Policy -> Maximum password age

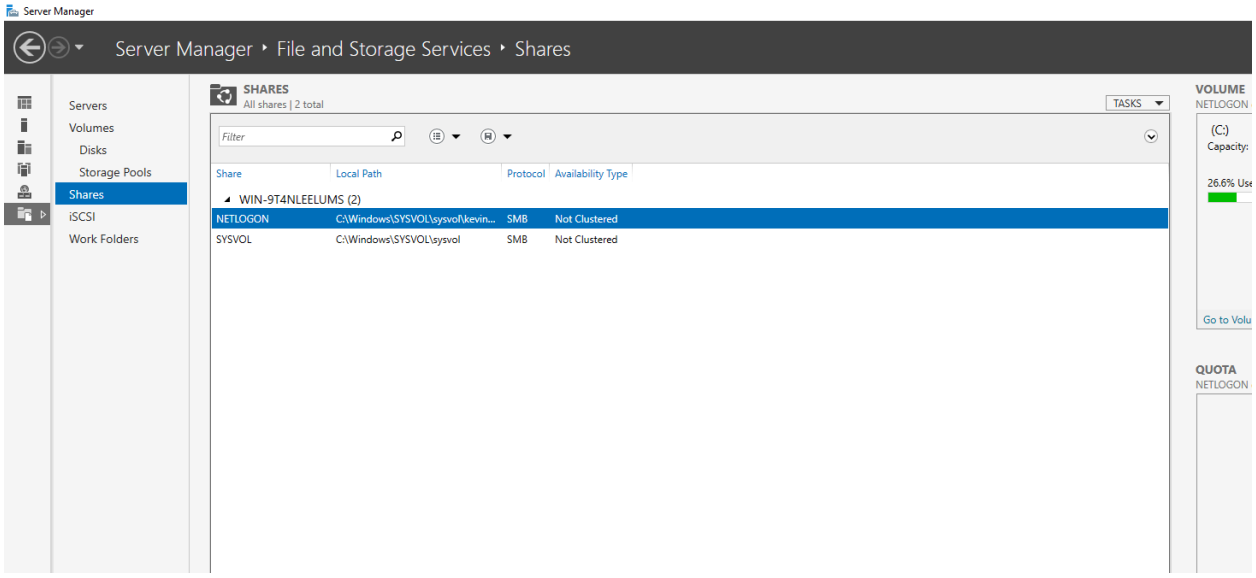


Or I want to set the login attempts to a certain time before the account is locked. I could do that too with the account lockout policy.



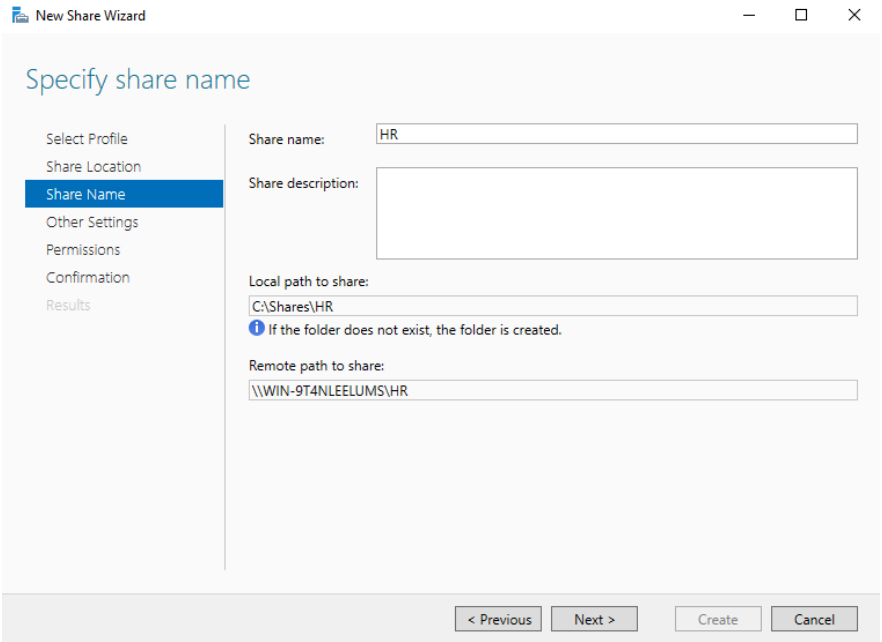
5. **Shared Folders and Permission Management**

- Opened **Server Manager**
- File and Storage Services → Shares



Let's say I want to share the HR folder with John Smith, since he will be joining the HR team. Currently there is no HR folder so I will create one.

Created a new folder called HR.

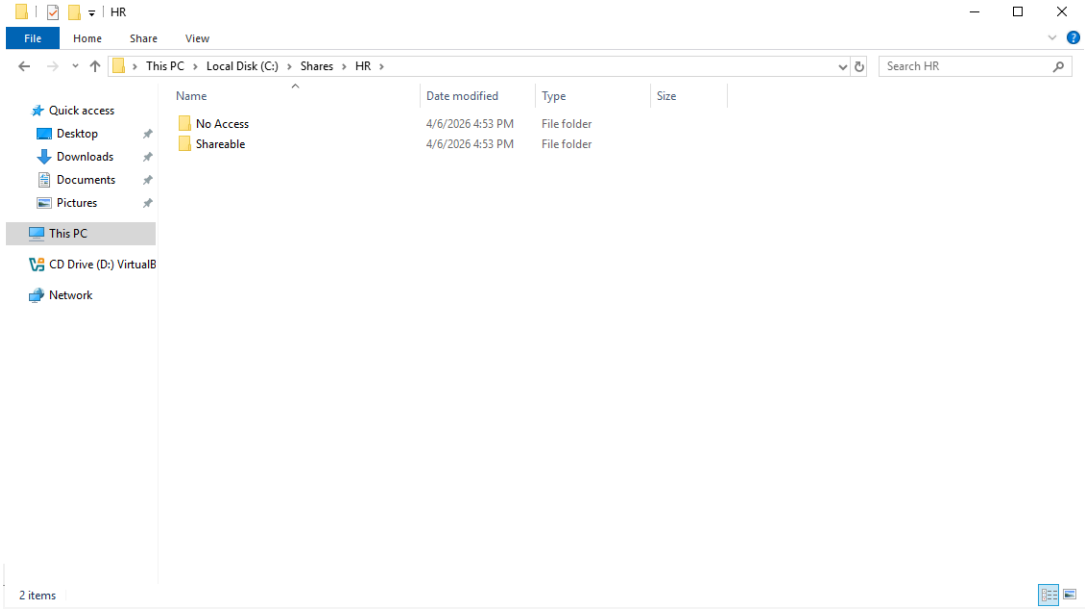


Within the HR folder, two subfolders were created:

- Shareable
- Private

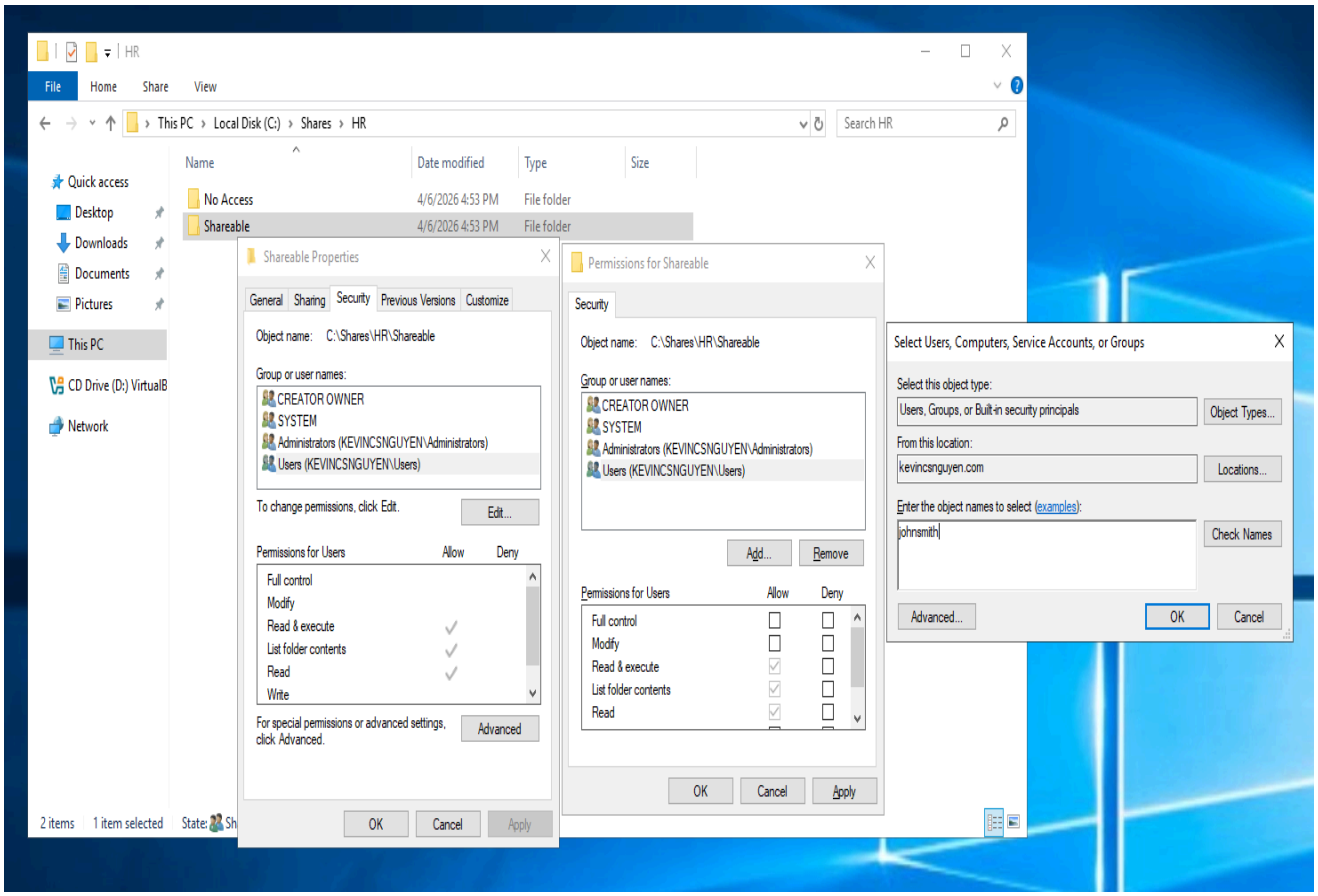
The **Shareable** folder was configured to allow access for the user **John Smith**.

I created 2 sub folders in the HR folder. Let's say I only want to share the "shareable" folder to John.

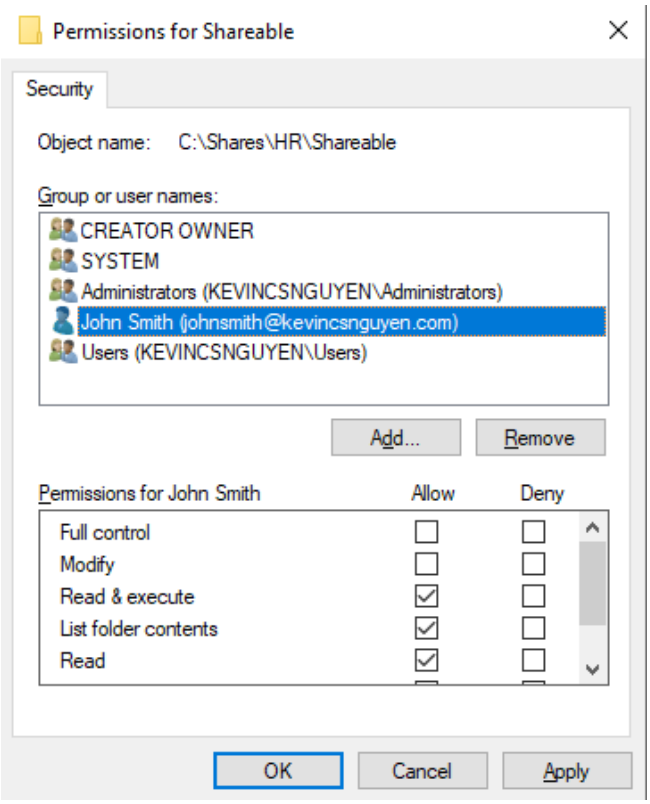
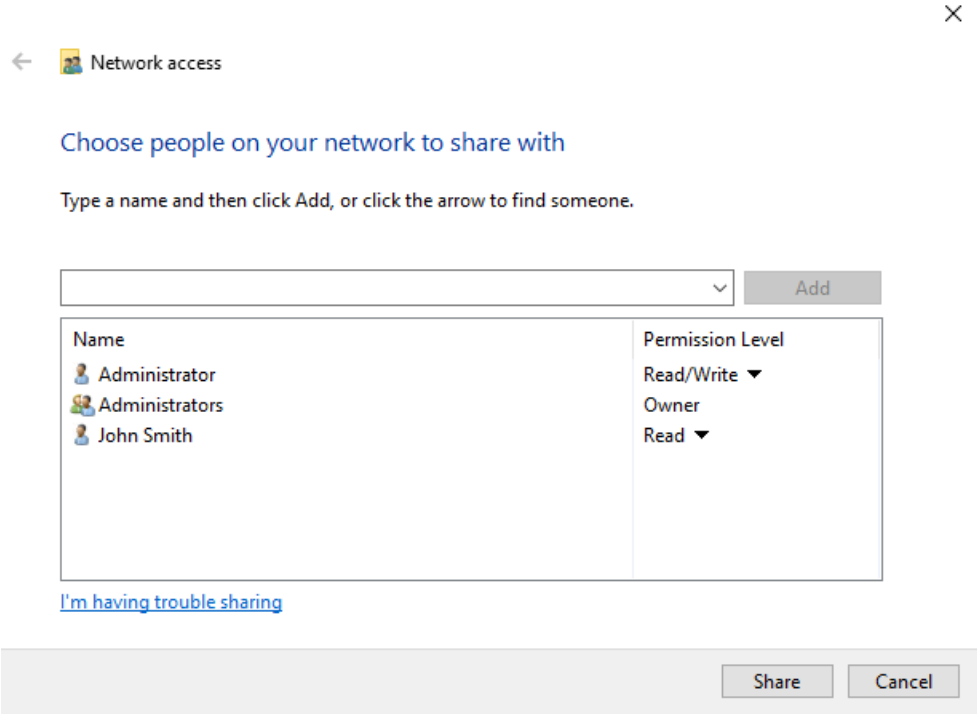


Permission configuration

- Right-click folder → **Properties**
- Navigate to **Security**
- Click **Edit** → **Add**
- Add user **johnsmith**



Permissions were then configured depending on access requirements.



John is successfully shared.